

# The $\sigma$ -Table Submatrix for the Müntz–Padé Spectral-Tau Solution

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## 1 Equation and Parameters

The constant-coefficient (time-independent) fractional Riccati equation:

$$D^\mu y(t) = P(v) + Q(v) \cdot y(t) + R(v) \cdot y(t)^2, \quad y(0) = 0$$

with parameters  $\mu = 1$ ,  $P = 1$ ,  $Q = 0$ ,  $R = -1$  (default: tanh oracle), working precision 128 bits ( $\approx 38$  decimal digits),  $N = 8$  orthogonal polynomials.

Discriminant:  $\Delta = Q^2 - 4PR = 4$ .

## 2 Program Output

### 2.1 Coefficient Functions

```
P(v) = 1    (constant)
Q(v) = 0    (constant)
R(v) = -1   (constant)
```

```
p(v) = 1
q(v) =
r(v) = -1
```

### 2.2 $\sigma$ -Table Definitions (Compiled Expressions)

```
a(k)  = a:when(k=1, p/Γ(+1),
               else, (Γ(((k-1)*)+1)/Γ((k* )+1))
               * ((q*a(k-1)) + (r*Σj a(j)*a((k-1)-j){j=1...k-2})))
m(k)   = m:k a(k+1)
(j)(k) = :j when(j=0, m(k),
               j=1, ((0)(k+1))-((0)*((0)(k))),
               else, (((j-1)(k+1))-((j-1)*((j-1)(k))))-((j-1)*((j-2)(k))))
h(j)   = h:j (j)(j)
(j)    = :j when(j=0, ((0)(1))/h(0),
               else, (((j)(j+1))/h(j))-(((j-1)(j))/h(j-1)))
(j)    = :j when(j=0, 0, else, h(j)/h(j-1))
Pn(n)  = Pn:n when(n=0, 0, n=1, hv(0),
               else, ((z- v(n-1))*Pn(n-1))- (v(n-1)*Pn(n-2)))
```

### 2.3 Jacobi Coefficients $\alpha_n, \beta_n, h_n$

| $n$ | $\alpha_n$ | $\beta_n$                | $h_n$                    | $\Delta\beta_n$    | $0 \in \Delta\beta_n$ |
|-----|------------|--------------------------|--------------------------|--------------------|-----------------------|
| 0   | 0          | 0                        | 1                        | —                  | —                     |
| 1   | 0          | $-0.3\overline{3}$       | $-0.3\overline{3}$       | $-0.3\overline{3}$ | false                 |
| 2   | 0          | $-0.0\overline{6}$       | $0.02\overline{2}$       | $0.2\overline{6}$  | false                 |
| 3   | 0          | $-0.02857\dots$          | $-6.349 \times 10^{-4}$  | $0.03810\dots$     | false                 |
| 4   | 0          | $-0.01587\dots$          | $1.008 \times 10^{-5}$   | $0.01270\dots$     | false                 |
| 5   | 0          | $-0.00\overline{1}$      | $-1.018 \times 10^{-7}$  | $0.005772\dots$    | false                 |
| 6   | 0          | $-0.00\overline{699300}$ | $7.119 \times 10^{-10}$  | $0.003108\dots$    | false                 |
| 7   | 0          | $-0.005128\dots$         | $-3.651 \times 10^{-12}$ | $0.001865\dots$    | false                 |

### 2.4 $\sigma(j, k)$ Submatrix ( $8 \times 8$ )

| $j \backslash k$ | 0           | 1                  | 2                  | 3                       | 4                      | 5                       | 6                       | 7                        |
|------------------|-------------|--------------------|--------------------|-------------------------|------------------------|-------------------------|-------------------------|--------------------------|
| 0                | 1           | $\emptyset$        | $-0.3\overline{3}$ | $\emptyset$             | $0.1\overline{3}$      | $\emptyset$             | $-0.05397$              | $\emptyset$              |
| 1                | $\emptyset$ | $-0.3\overline{3}$ | $\emptyset$        | $0.1\overline{3}$       | $\emptyset$            | $-0.05397$              | $\emptyset$             | $0.02187$                |
| 2                | 0           | $\emptyset$        | $0.02\overline{2}$ | $\emptyset$             | $-0.009524$            | $\emptyset$             | $0.003880$              | $\emptyset$              |
| 3                | $\emptyset$ | 0                  | $\emptyset$        | $-6.349 \times 10^{-4}$ | $\emptyset$            | $2.822 \times 10^{-4}$  | $\emptyset$             | $-1.154 \times 10^{-4}$  |
| 4                | 0           | $\emptyset$        | 0                  | $\emptyset$             | $1.008 \times 10^{-5}$ | $\emptyset$             | $-4.581 \times 10^{-6}$ | $\emptyset$              |
| 5                | $\emptyset$ | 0                  | $\emptyset$        | 0                       | $\emptyset$            | $-1.018 \times 10^{-7}$ | $\emptyset$             | $4.698 \times 10^{-8}$   |
| 6                | 0           | $\emptyset$        | 0                  | $\emptyset$             | 0                      | $\emptyset$             | $7.119 \times 10^{-10}$ | $\emptyset$              |
| 7                | $\emptyset$ | 0                  | $\emptyset$        | 0                       | $\emptyset$            | 0                       | $\emptyset$             | $-3.651 \times 10^{-12}$ |

## 3 Analysis

### 3.1 Checkerboard Sparsity

$\sigma(j, k) = 0$  whenever  $j + k$  is odd. The moment sequence  $m(k) = a(k + 1)$  for  $\tanh(t)$  has every other entry zero because  $\tanh$  is an odd function and only odd-power Taylor coefficients survive:

$$\tanh(t) = t - \frac{1}{3}t^3 + \frac{2}{15}t^5 - \frac{17}{315}t^7 + \dots$$

so  $a(2) = a(4) = a(6) = \dots = 0$ , and the Chebyshev–Wheeler recurrence preserves this parity structure.

### 3.2 Diagonal Equals $h(j)$

The diagonal entries  $\sigma(j, j) = h(j)$  match the  $h_n$  column of the Jacobi coefficient table exactly. These are the squared  $L^2$ -norms  $\|P_j\|^2$  of the  $j$ -th orthogonal polynomial under the moment functional.

### 3.3 Upper-Left Triangular Fill with Super-Exponential Decay

Nonzero entries concentrate in the upper-left triangle and decay super-exponentially along the diagonal:

$$1 \rightarrow 0.022 \rightarrow 10^{-5} \rightarrow -10^{-7} \rightarrow 7 \times 10^{-10} \rightarrow -4 \times 10^{-12}$$

This is the moment sequence being compressed into Jacobi coefficients. Each diagonal entry  $h(j)$  shrinks by roughly two orders of magnitude per step, reflecting the rapid convergence of the tridiagonalization.

### 3.4 Zeros Accumulate Below the Anti-Diagonal

By row 6–7, all entries below the anti-diagonal are numerically zero. This is the Lanczos/Danilevsky reduction reaching its fixed point: the new row  $\sigma(j)$  becomes a linear combination of the previous two rows with constant coefficients  $\alpha, \beta$ . Exactly at this point  $0 \in \Delta\beta_n$  triggers autotermination in the Jacobi table above.

### 3.5 Columns Beyond $j + 1$ Decay Rapidly

For fixed row  $j$ , the entries  $\sigma(j, j+2), \sigma(j, j+4), \dots$  are the “remainder” moments not yet absorbed into the tridiagonal form. Their rapid shrinkage is why  $\beta_n$  converges so fast for this equation—the moment information is exhausted within a few rows.